



How infants predict respect-based power

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ABSTRACT

Research has shown that infants represent legitimate leadership and predict continued obedience to authority, but which cues they use to do so remains unknown. Across eight pre-registered experiments varying the cue provided, we tested if Norwegian 21-month-olds ($N=128$) expected three protagonists to obey a character even in her absence. We assessed whether bowing for the character, receiving a tribute from or conferring a benefit to the protagonists, imposing a cost on them (forcefully taking a resource or hitting them), or relative physical size were used as cues to generate the expectation of continued obedience that marks legitimate leadership. Whereas bowing sufficed in generating such an expectation, we found positive Bayesian evidence that all the other cues did not. Norwegian infants unlikely have witnessed bowing in their everyday life. Hence, bowing/prostration as cue for continued obedience may form part of an early-developing capacity to represent leadership built by evolution.

1. Introduction

Different forms of social hierarchies permeate the human life world and transitive rank has deep evolutionary roots in the dominance hierarchies found among other species (Boehm, 1999; Fiske, 1991; Kaufmann, 1983; Sapolsky, 2004; Sidanius & Pratto, 1999). The ability of the formidable to inflict costs upon their opponents makes it adaptive to yield from fights one is unlikely to win (Maynard-Smith, 1982). Hence, core representations and motives for dominance rank could evolve (Chen Zeng et al., 2022; Thomsen & Carey, 2013; see also Thomsen, 2020). Consistent with this, even preverbal infants expect the more formidable agents to prevail in zero-sum conflict (Gazes et al., 2017; Mascaro & Csibra, 2012; Mascaro et al., 2023; Pun et al., 2016; Thomsen et al., 2011), toddlers prefer those for whom others yield in such conflicts (Thomas et al., 2018), and bonobos prefer novel third parties who dominate and hinder others (Krupenye & Hare, 2018). However many, and perhaps most, human hierarchies are not based on brute dominance, but instead on legitimate and consensual leadership, authority and prestige (Cheng, 2020; Fiske, 1992; Henrich & Gil-White, 2001; McClanahan et al., 2022). Here, authorities are expected to look out for their followers (Fiske, 1991), prestige is thought to reflect the competence and benefits one may share (Henrich & Gil-White, 2001), and respecting leadership is seen as morally good (Curry et al., 2019; Rai & Fiske, 2011).

Legitimate leadership, and motives to adopt leader–follower relations, might have evolved in humans because they help solve critical coordination and collective action problems – such as conflict resolution and resource management within the group, as well as coalitional defense against rival groups – thus allowing individuals belonging to a group to reap the benefits of cooperation (Ahmad &

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Loch, 2020; Garfield et al., 2020; Glowacki & von Rueden, 2015; Pietraszewski, 2020; Van Vugt & von Rueden, 2020). To be sure, leadership and followership are ubiquitous and central phenomena in human social groups (Van Vugt & Ahuja, 2011). Leadership can be observed in large-scale hierarchical societies as well as in small-scale relatively egalitarian societies, though in a less institutionalized and more situational form (Garfield et al., 2019, 2020; Lewis, 1974; von Rueden & Van Vugt, 2015). To some extent, leadership can even be observed among non-human animals (Couzin et al., 2005; Smith et al., 2016). If the evolution of leadership and followership played a crucial role for sustaining human cooperation and collective coordination, aspects at the heart of group living, then our mind may be endowed with core, reliably and early-developing mechanisms for representing and respecting legitimate leadership as well as dominance, so that human infants need not only live in a *Hobbesian* 'state of nature' where might makes right.

Consistent with this proposal, recent research has demonstrated that within their second year of life, just-linguistic infants distinguish between respect-based social power/legitimate leadership, and fear/coercion-based power (for the distinction, see French et al., 1959) and that they expect leaders to work for the better of the group. In particular, though infants prefer those for whom others voluntarily defer, this does not include those who prevail by brute force (Thomas et al., 2018). In fact, infants expect leaders to bear the cost of intervening to rectify social transgressions within the group (Stavans & Baillargeon, 2019). Just-linguistic infants also expect a group of agents to continue to comply with an order issued by a leader (respect-based power) even after she leaves the group alone, but they do not hold the same expectation when the individual issuing the order is a bully (fear-based power) (Margoni et al., 2018). Yet, it remains unknown which cues are necessary and sufficient for identifying a legitimate leader in the infant mind. Knowing which features are critical for predicting whose directions will be respected may reveal core aspects of a fundamental, dedicated and early-developing capacity to represent legitimate leadership and authority.

Here, we addressed this question across eight experiments with 128 Norwegian 21-month-old infants. We adapted the paradigm of Margoni et al. (2018) which demonstrated that Italian and North American infants expected a group to comply with an order to go to bed issued by a bully (who had hit them with a stick and taken their ball), but only when she remained on the scene, watching and controlling the group, and not once she left the scene. In contrast, when the same order was issued by a leader carrying a stick (for whom the group had bowed before offering their ball and who bowed slightly in return), then expectations of continued obedience were generated, i.e., even in her absence. This suggested that infants understood that the leader's directive power was based on respect and voluntary deferral to her directions, even when she was not around to enforce obedience, rather than on immediate controlling coercion. In the present research, we systematically varied the series of candidate cues listed below to test which, if any, sufficed in eliciting the expectation of continued obedience (staying in bed).

Bowing, or prostration and lowered/shrinking body posture, not only features in the respect and deference displays found across cultural practices and religious rituals (Fiske, 1991, 1992, 2004; Fiske et al., 2009; Mallory, 1891), but also in the submission displays found across species, including primates such as chimpanzees, baboons, and macaques; elephants; wolves and dogs; seals; rabbits; crayfish; and salamanders (de Waal, 1989; Maslow, 1936; Tracy & Matsumoto, 2008). Conversely, status, success, winning, and dominance displays across cultures (Fiske, 1991, 1992, 2004; Fiske et al., 2009; Tracy & Matsumoto, 2008) and species (Brown & Maurer, 1986; Buston, 2003; Clutton-Brock et al., 2006) involve upright/expansive body postures (Witkower et al., 2023). These dimensions of above-below and bigger-smaller are also found in conceptual metaphors for social rank across language families (Lakoff & Johnson, 1980), and indeed even preverbal infants expect the taller, bigger figures to prevail in zero-sum conflict (Thomsen et al., 2011; see also Thomsen & Carey, 2013). This begs the question if bowing/prostration signals respect for a legitimate leader and authority, submission to a dominant other, or both, in the human and infant mind. Note that both Thomsen et al. (2011) and Margoni et al. (2018) used bowing/prostration in their infant paradigms to signal deference to a dominant other or a leader, respectively. The original right-of-way dominance paradigm (Thomsen et al., 2011), as well as follow-ups utilizing it (Pun et al., 2016, 2021; Thomas et al., 2018) confound bowing and yielding in zero-sum conflict. However, subsequent work has found that preverbal infants do not simply link bowing, rather than yielding, to relative formidability (Thomsen & Carey, 2012), and that toddlers do *not* prefer those for whom others simply bow although they *do* prefer novel agents for whom others bow and yield the way (Thomas et al., 2018). Thus, bowing does not suffice in generating dominance/formidability-linked predictions and motives in the absence of information about conflict outcomes. Here, we test if bowing might instead be a dedicated cue for respect-based leadership so that it would suffice in generating predictions of continued obedience to an absent individual's prior directions.

Conferring benefits. Competence-based models of leadership hold that the respect leaders enjoy and their attainment of social status, authority and prestige follow from their ability and willingness to generate benefits for the group (Cheng et al., 2013; Durkee et al., 2020; Henrich, 2017; Henrich & Gil-White, 2001; see also Fiske, 1991, 1992). Indeed, evidence from small-scale societies indicates that generosity displays (if not presumptuous; Boehm, 1999) and commitment to the group are associated with leadership and social rank (Anderson & Kilduff, 2009; Hardy & Van Vugt, 2006; von Rueden et al., 2014; see also Fiske, 1991). If conferring benefits upon followers is key for the respect and directive power that leaders enjoy, then it might suffice in generating expectations for continued obedience even in the infant mind.

Relative size. Still, leadership inferences from conferring benefits to the group may only be generated if it is obvious that one was conferring, rather than simply losing, benefits to others: Because even preverbal infants associate losing resources with having lower dominance rank (Gazes et al., 2017; Mascaro & Csibra, 2012; see also Enright et al., 2017; Hok et al., 2024), perhaps it must be clear that leaders do have the highest resource-holding potential (Parker, 1974) when *choosing* to confer benefits upon their followers. Because even preverbal infants associate relative size with such social dominance (Thomsen et al., 2011), we also combined bestowing a benefit (a ball) upon the group with having greater physical size. Note, however, that respected authorities, high priests, and religious deities are also represented as being bigger and higher across cultures (Fiske, 1991, 1992, 2004); that taller people are more likely to be chosen as leaders (Lindqvist, 2012; Schubert et al., 2008; Stogdill, 1948) and even presidents (Persico et al., 2004); and that the height of a leader also relates to adult perceptions of their intelligence and charisma, as well as their perceived dominance (Blaker et al., 2013;

Blaker & Van Vugt, 2014; Hamstra, 2014). Hence, in principle relative physical size might suffice in cueing respected leadership in the infant mind.

Receiving tribute. Another core aspect of leadership, observed in large- and small-scale societies, is that leaders get compensated for the costs they incur in guiding and coordinating the group, receiving material, social and reproductive benefits in return (Garfield et al., 2020; Hooper et al., 2010). Here we tested if receiving tribute (e.g., the fact that the group offers the main character their ball) suffices in generating the representation of a respected leader whose order will be continuously obeyed. Note that Margoni et al. (2018) included a control condition (i.e., *powerless-character condition*) in which infants were presented with a main character who greeted and was greeted by the group in a friendly manner before receiving their ball in tribute, and that in this condition infants did *not* generate the expectation of continued obedience. Possibly, this was because the infant participants inferred that all the characters were friends who share things but need not follow each other's orders. Given the centrality of receiving tribute in models and observations of leader–follower relations, here we tested if receiving tribute alone (i.e., without the presence of friendly exchange between agents) solicits inferences of continued obedience.

Imposing Costs & Violent Intimidation. Finally, and importantly, Margoni et al. (2018)'s finding that physical, violent intimidation combined with confiscating resources does *not* solicit infant predictions of (continued) obedience bears replication because it further qualifies our understanding of the core relational underpinnings of leadership in the human mind: Conflict-based models (Mazur, 1973; Smith & Parker, 1976) posit that a core dimension of dominance rank is the ability (and willingness) to inflict cost to coerce or intimidate others to get your way. Not only do infants represent social dominance (Mascaro & Csibra, 2012; Thomsen et al., 2011), expecting for instance that more physically dominant agents will get a larger shares (Enright et al., 2017), but preschoolers also link malevolent actions and indifference to others' needs to social power, and more so than they do benevolent actions to benefit others (Gülgöz & Gelman, 2017; Terrizzi et al., 2020; Thomas et al., 2022; Wang et al., 2024; see also Thomsen, 2020). Note also that dual-pathway models suggest that cost infliction (or dominance) and benefit generation (or prestige granted to individuals whose expertise benefits the group) are both possible, but independent ways to the top (Cheng et al., 2013; Henrich & Gil-White, 2001). Interestingly, in situations where free-riding severely threatens the existence of the group, as during collective defense and wartime, the link between dominance and leadership strengthens (Laustsen & Petersen, 2017; von Rueden & Van Vugt, 2015; Van Vugt & Spisak, 2008). Thus, even if leaders would not emerge by simply intimidating others (Boehm, 1999; see also Ronay et al., 2023), formidability cues of resource holding potential, physical size, strength and aggression might be used in the selection of leaders (Blaker & Van Vugt, 2014; Lukaszewski et al., 2016). Hence, it is theoretically possible that core predictions specify that individuals will obey those who can hurt or impose cost on them, so that replicating Margoni et al. (2018) that this is not a core prediction made by human infants is important.

Finally, note that imposing a cost, even if backed by formidability-based coercion as in a winning physical zero-sum resource conflict, does not equate violent, out-of-the-blue, bullying aggression. Nevertheless, such "random acts of aggression and senseless acts of intimidation" appear to bolster social status in non-human primate groups (Silk, 2002; see also Krupenye & Hare, 2018) because of the fear and submission they inflict. Hence, here we also separate the two cues of the bully-condition of Margoni et al. (2018) – namely claiming a resource (ball) through physical coercion versus attacking and violently hitting others (with a stick) – to replicate if neither generates expectations of continued obedience among human infants.

In sum, here we tested which of the cues above-mentioned are so fundamental and unique to respect-based power that they would signal leadership even to the infant mind. We investigated this in the egalitarian cultural setting of Norway where clear cues to social rank, such as bowing, are rarely displayed. Across experiments, Norwegian infants were shown one cue in the familiarization phase, and the obey and disobey events in the test phase. Using the well-established Violation-of-Expectation paradigm (see Margoni et al., 2024), which has been widely used to test both human infant cognition and non-human animal cognition (Ginnobili & Olmos, 2021), we assessed which cues generated prolonged looking at disobedience, thus indicating the expectation of continued obedience and the representation of respect-based power.

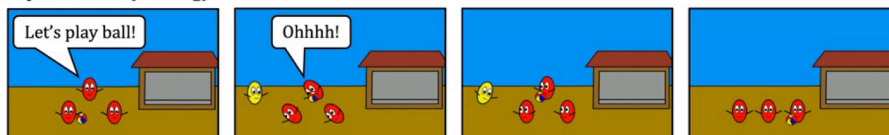
The raw data and Supplementary Material (SM), including Movies S1–S14, are available at OSF, 2023, link: <https://osf.io/5mbpv/>. The experiments protocol, predictions, inclusion/exclusion criteria and analyses were pre-registered (link: <https://aspredicted.org/yg58c.pdf>).

2. Experiment 1 – Bowing

We tested if bowing (the fact that the group initially bowed before the character who then gave them an order) sufficed in generating the representation of a leader whose directions will be obeyed and respected even in her absence. Infants were first presented with a pair of character-familiarization trials where they saw a computer-animated event where three protagonists (red oval shapes with faces, stick arms and female voices) bowed before a character (a yellow oval) in unison while saying "Ohhhh!" in a reverential tone; to that, the character replied with a hint of a bow (see Movie-S1, Fig. 1). Next, infants were presented with a pair of order-familiarization trials aimed at familiarizing them to the character's order: The character stretched her left arm pointing at the house and said, "Time for bed!" (in Norwegian, "Nå er det leggetid!") (see Movie-S2 and Fig. 2). Last, infants were presented with test

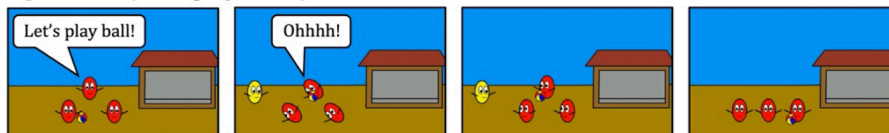
Character-Familiarization Trials

A Experiment 1 (Bowing)



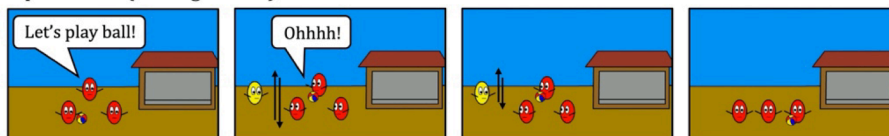
The protagonists bow 2 times, and the character responds with a hint of a bow

B Experiment 2 (Bowing replication)



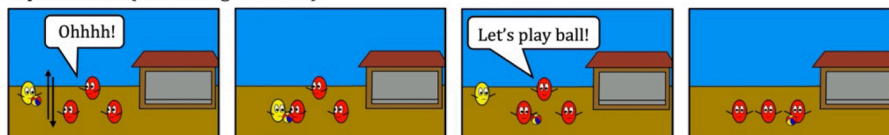
The protagonists bow 2 times, and the character stands still

C Experiment 3 (Bowing control)



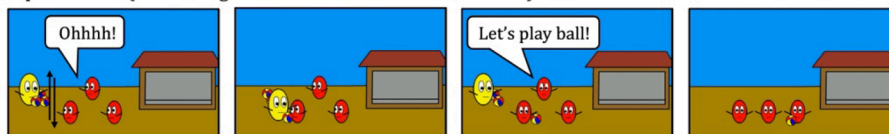
The protagonists move up-and-down

D Experiment 4 (Conferring a benefit)



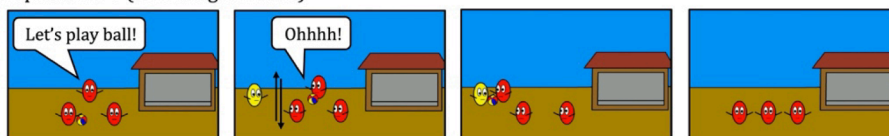
The protagonists move up-and-down, and the character gives them the ball

E Experiment 5 (Conferring a benefit and size of the character)



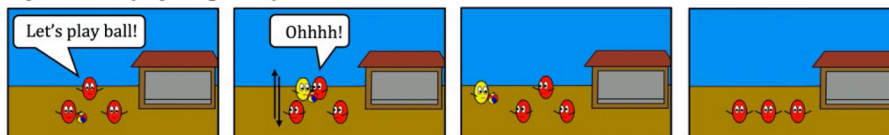
The character, now bigger in size, gives the protagonists one ball

F Experiment 6 (Receiving a tribute)



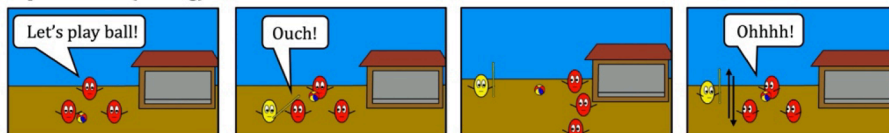
The protagonists give the character their ball

G Experiment 7 (Imposing a cost)



The character takes the protagonists' ball

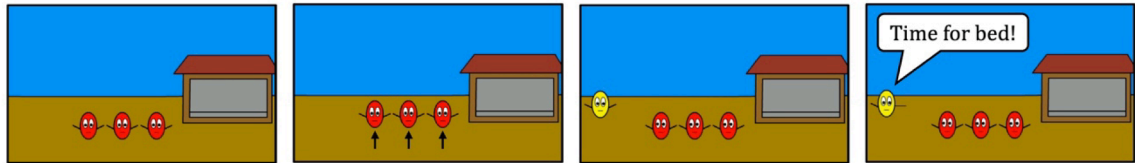
H Experiment 8 (Hitting)



The character hits the protagonists with a stick

Fig. 1. Schematic Depiction of the Character-Familiarization Trials in Experiments 1 to 8.

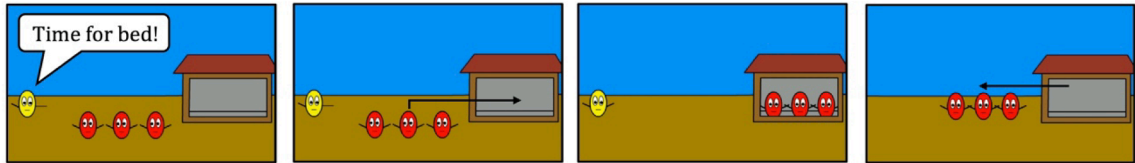
Order-Familiarization Trials



The protagonists jump 3 times

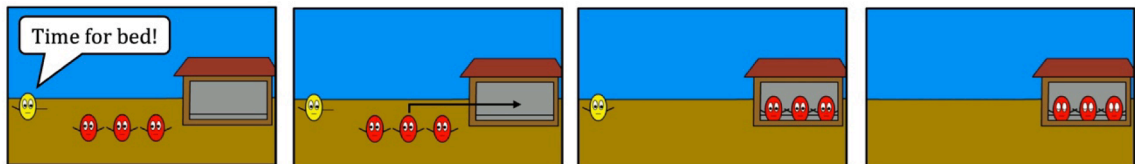
Test Trials

A Disobey Event



The protagonists disobey once the character issuing the order leaves

B Obey Event



The protagonists continue obeying when the character issuing the order leaves

Fig. 2. Schematic Depiction of the Order-Familiarization and Test Trials in Experiments 1 to 8.

trials showing the character again issuing the order and the protagonists complying and filing into their house on the left of the scene; once the character left the scene, either the protagonists continued to comply and close their eyes, as though going to sleep (*obey event*; Movie-S3), or they disobeyed leaving the house and filing back out into the field (*disobey event*; Movie-S4) (see Fig. 2).

2.1. Methods

2.1.1. Participants

For all the experiments (1 to 8), we determined the sample size using a Sequential Bayes Factors approach (see Schönbrodt et al., 2017). We continued collecting data until we reached at least positive and moderate Bayesian evidence (cutoff, Bayes Factor ≥ 3) for either the alternative hypothesis of longer looking at either one of the two test events or the null hypothesis of equal looking at the two test events (for a Bayesian paired samples *t*-test). At least 16 participants were recruited. Parents provided informed consent, and the procedure we employed was approved by the Norwegian Centre for Research Data (NSD).

Participants were 16 full-term infants (5 female; $M_{\text{Age}} = 21$ mo, 12 d; age range = 19 mo, 16 d to 23 mo, 29 d) from Norwegian-speaking families. Another 12 infants were tested but excluded because they looked the maximum time allowed in both test pairs (4), because they were overly fussy or active (4) or distracted in test by food items (1), or because of parental interference in test (1) or bad internet connection (2). Attrition rate was especially high in Exp. 1 and varied to some extent across the remaining experiments (from 4 to 10 excluded infants, with $M=5.86$, $SD=2.34$). This variability might be attributed to chance alone, but the high rate in Exp. 1 could perhaps also stem from the fact that this was the first experiment conducted on-line by the main experimenter (the first author).

2.1.2. Design

Infants were presented with one pair of character-familiarization trials followed by one pair of order-familiarization trails followed by two pairs of test trials. Each character-familiarization trial (Fig. 1A) presented infants with a 24-s computer-animated event in

which oval shapes with faces, stick arms and female voices interacted with each other. This event introduced three protagonists and a character, and provided infants with the *bowing* cue. To start, the three protagonists (red ovals) stood at the center of the scene, in an open field, to the left of a house with a large front opening; they said, “Let’s play ball!” (in Norwegian, “La oss kaste ball!”), and passed a colorful ball to each other until the lone character (a yellow oval) entered the scene from the left. Once the character entered the scene and stopped in front of the group, the protagonists bowed before her in unison while saying “Ohhhh!” in a reverential tone; to that, the character replied with a hint of a bow. The bowing sequence was repeated two times, then the character left the scene, and the protagonists formed a line (Movie-S1). This event repeated, after a blank screen, until the trial ended (see Procedure for criteria).

Next, each order-familiarization trial (Fig. 2) presented infants with a 7-s event that had the purpose of familiarizing infants with the yellow character’s order. The protagonists stood in a line at the center of the stage and jumped together three times until the character arrived, stretched her left arm pointing at the house and said, “Time for bed!” (“Nå er det leggetid!”) (Movie-S2). This event repeated until the trial ended. Last, each test trial presented infants with a 15-s event which began like the order-familiarization event but continued with the protagonists first complying with the order and filing into the house through its left side. Once filed into the house, they could be seen through its front opening. At that point, the character who issued the order left the scene and either the protagonists continued to comply and remained in the house closing their eyes, as though going to sleep (*obey event*; Movie-S3), or they disobeyed leaving the house and filing back out in the field (*disobey event*; Movie-S4) (Fig. 2). Each animation repeated until the trial ended. Test events were presented on alternate trials, whose order (first obey or first disobey) was counterbalanced across participants.

2.1.3. Procedure

Infants sat on a parent’s lap or on a baby chair next to their parent. Parents were instructed to close their eyes and not to interfere with their child’s behavior. We tested infants remotely at their home using *Zoom*, presenting our stimuli on the parent’s computer (Chuey et al., 2021). Looking behavior was monitored and coded on-line using *PyHab* (Kominsky, 2019), an add-on for *PsychoPy* (Peirce et al., 2019). On-line coding was done by the first author blind to the order of test trials (a cardboard placed over the relevant portion of the experimenter’s screen prevented them from knowing the content of the videos). We recorded the video-call, and captured the infant and the movies simultaneously. Additional details about the procedure we followed for testing infants on-line via *Zoom* are provided in the SM. All observations (16/16 participants) were recoded off-line by two independent coders, both blind to the content of the movies. Interobserver agreement averaged 97 % per trial per infant. We used the first observer’s responses in the analyses.

Infants received two pairs of familiarization trials (one pair of character-familiarization trials and one pair of order-familiarization trials) and two pairs of test trials. Each trial began with a bright attention getter (a smiling baby face). Each character-familiarization trial ended when infants (i) looked away for 2 consecutive seconds after having looked for at least 24 cumulative seconds (the duration of one event loop) or (ii) when 72 consecutive seconds had elapsed (the duration of three event loops). If the minimum on-time criterion of 24 s was not met, the trial was presented again. Each order-familiarization trial ended when infants (i) looked away for 2 consecutive seconds after having looked for at least 7 cumulative seconds (the duration of one event loop) or (ii) when 35 s had elapsed (the duration of five event loops). If the minimum on-time criterion was not met, the trial was presented again. Last, each test trial ended when infants (i) looked away for 2 consecutive seconds after having looked for at least 15 cumulative seconds (one event loop) or (ii) when 60 s had elapsed (the duration of four event loops). If the minimum on-time criterion of 15 s was not met, the same trial was presented again.

2.2. Results

Looking times data were log-transformed prior to the analyses (Csibra et al., 2016). Note, however, that for ease of communication, in the text and in the graphs we report raw looking times. A paired samples *t*-test revealed that infants looked significantly longer at the disobey ($M=42.69$, $SD=13.44$) than at the obey ($M=33.56$, $SD=13.95$) event following the familiarization trials where the group bowed repeatedly before the character who acknowledged this with a small bow, $t(15) = 2.62$, $p = 0.019$, Cohen’s $d = 0.65$, $BF_{+0} = 6.26$ (Fig. 3). A Bayes Factor (BF) of 6.26 (with a default Cauchy prior width of 0.707; JASP Team, 2022) indicates that the observed data are about 6 times more likely to occur under the alternative hypothesis (disobey > obey) than under the null hypothesis (disobey = obey). Further analyses testing the effect of age, order of test trials, and infant sex can be found in the SM.

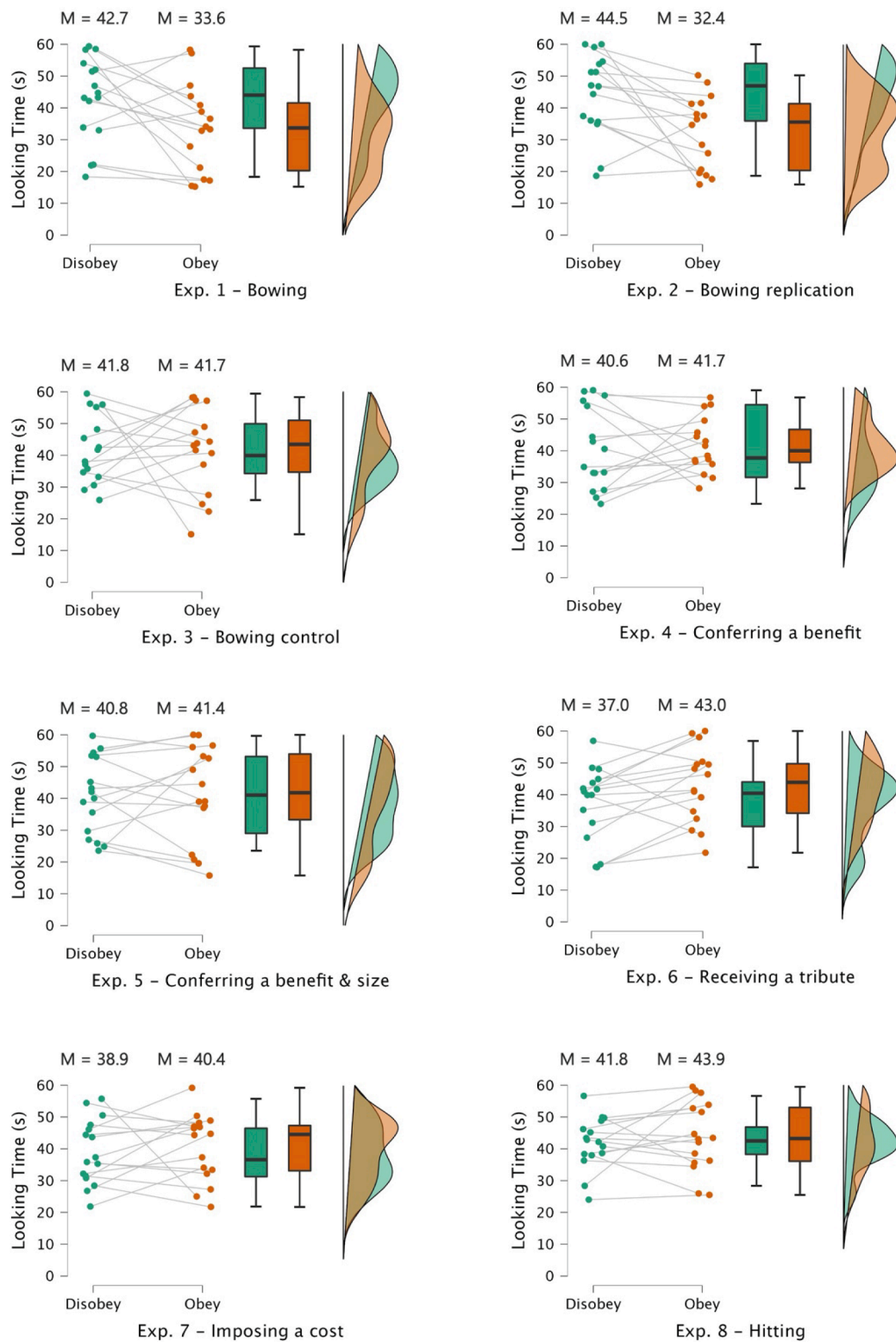


Fig. 3. Results of Experiments 1 to 8.

2.3. Interim discussion

In sum, infants looked reliably longer at disobedience than at obedience, suggesting that they expected the protagonists to spontaneously comply with the demands of an agent for whom they have bowed. Thus, bowing sufficed in generating the prediction of respect and continued obedience.

3. Experiment 2 – Bowing replication

We tested if we could replicate the result of Exp. 1 to show that bowing before a character suffices in generating the representation of her leadership, even when the character does not respond with a slight bow in return (as in Exp. 1 and Margoni et al., 2018). Familiarization and test events were thus identical to those of Exp. 1 except that in the character-familiarization event the character did not reply with a hint of a bow (Fig. 1B and Movie-S5).

3.1. Methods

3.1.1. Participants

Participants were 16 full-term infants (8 female; $M_{Age} = 21$ mo, 15 d; age range = 19 mo, 2 d to 23 mo, 8 d) from Norwegian-speaking families. Another 5 infants were excluded because overly fussy or active (3), because of parental interference (1) or bad internet connection (1).

3.1.2. Design

Familiarization and test events were identical to those of Exp. 1 with the exception that in the character-familiarization event the yellow character did not reply with a hint of a bow but stood still after the protagonists bowed for her (Fig. 1B and Movie-S5).

3.1.3. Procedure

The procedure was identical to that of Exp. 1 with the exception that we added a pair of test trials. Interobserver agreement averaged 96 % per trial per infant.

3.2. Results

Infants looked significantly longer at the disobey ($M=44.48$, $SD=12.91$) than at the obey ($M=32.37$, $SD=11.47$) event, $t(15) = 2.96$, $p = 0.010$, $d = 0.74$, $BF_{+0} = 11.04$ (Fig. 3); $BF_{+0} = 160.10$ if data from Exp. 1 and 2 are pooled together (whereas a $BF=11$ indicates strong evidence, a $BF=160$ indicates extremely strong evidence).

3.3. Interim discussion

In sum, we replicated the finding of Exp. 1 and found strong evidence that Norwegian just-linguistic infants use bowing as a cue for leadership, even though they rarely observe bowing in everyday life. We further demonstrated that the expectation of continued obedience does not require that the agent who receives the bowing responds to it.

4. Experiment 3 – Bowing control

With Exp. 3 we tested an alternative low-level explanation of the findings in Exp. 1 and 2, namely that the expectation of obedience was not generated in response to the act of bowing itself, but to the simple fact that the protagonists performed a coordinated movement in front of the character. We used the same events used in Exp. 1 except that the protagonists no longer bowed, they instead moved up-and-down in unison while saying “Ohhhh!”, and the character replied with a similar movement (Fig. 1C and Movie-S6).

4.1. Methods

4.1.1. Participants

Participants were 16 full-term infants (10 female; $M_{Age} = 21$ mo, 9 d; age range = 19 mo, 17 d to 23 mo, 14 d) from Norwegian-speaking families. Another 6 infants were excluded because they were overly fussy or active.

4.1.2. Design

Events were identical to those of Exp. 1 except that in the character-familiarization event the protagonists did not bow, they instead moved up-and-down in unison while saying “Ohhhh!”, and the character replied with a similar movement (Fig. 1C and Movie-S6).

4.1.3. Procedure

The procedure was identical to that of Exp. 2, except that for experiments such as Exp. 3 where a main null result was revealed, we randomly selected 25 % of the total observations (4/16 infants) to recode off-line. Interobserver agreement averaged 98 % per trial per infant.

4.2. Results

Infants looked about equally at the disobey ($M=41.82$, $SD=10.62$) and obey ($M=41.70$, $SD=13.48$) events, $t(15) = 0.28$, $p = 0.785$, $d = 0.07$, $BF_{0+} = 3.14$ (Fig. 3). Here, a BF of 3.14 provides positive evidence for the null hypothesis, indicating that the observed data are about three times more likely to occur under the null hypothesis (disobey = obey) than under the alternative hypothesis (disobey > obey). Last, looking times during the test events (disobey, obey) were compared using an ANOVA with experiment (Exp. 2, Exp. 3) as between-subjects factor. As predicted, this analysis revealed a significant interaction between test event and experiment, $F(1, 30) = 4.23$, $p = 0.049$, $\eta_p^2 = 0.12$ (see the SM).

4.3. Interim discussion

In sum, in Exp. 3 infants made no predictions whether the group would obey a character before whom they had moved in a coordinated manner. Thus, infants in Exp. 1 and 2 specifically reacted to the display of spontaneous deference or respect signified by the groups bowing before the character.

5. Experiment 4 – Conferring a benefit

We tested if conferring benefit (the fact that the character gave the group a ball) sufficed in generating the prediction of continued obedience. Infants were presented with a character-familiarization event where the character conferred a benefit to the group, giving them a ball (Fig. 1D, Movie-S7).

5.1. Methods

5.1.1. Participants

Participants were 16 full-term infants (8 female; $M_{Age} = 21$ mo, 2 d; age range = 19 mo, 5 d to 23 mo, 16 d) from Norwegian-speaking families. Another 4 infants were excluded because they were overly fussy or active.

5.1.2. Design

Events were identical to those of Exp. 3 except that in the character-familiarization event the character conferred a benefit to the group, giving them a ball to play. Each character-familiarization trial (Fig. 1D) presented infants with a 29-s event. To start, the protagonists stood at the center of the scene without a ball until the character entered the scene from the left with a colorful ball. Once the character entered the scene and stopped in front of the group, the same sequence used in Exp. 3 where the protagonists moved up-and-down was presented (except that the character stood still). Next, the character approached the group, gave the ball to the closest protagonist, and returned to her position. The three protagonists proceeded to play ball like in Exp. 1, the character left the scene and the protagonists formed a line (Movie-S7).

5.1.3. Procedure

The procedure was identical to that of Exp. 3 except that each character-familiarization trial ended when infants (i) looked away for 2 consecutive seconds after having looked for at least 29 cumulative seconds or (ii) when 87 s had elapsed (the duration of three event loops). Interobserver agreement averaged 98 % per trial per infant.

5.2. Results

We found positive Bayesian evidence that infants looked about equally at the disobey ($M=40.64$, $SD=12.84$) and obey ($M=41.68$, $SD=8.64$) events, $t(15) = 0.61$, $p = 0.554$, $d = 0.15$, $BF_{0+} = 5.78$ (Fig. 3).

5.3. Interim discussion

This result suggests that conferring benefit does *not* suffice in generating the prediction that one's directions will be followed. We followed up Exp. 4 by testing if the prediction could be generated when the character conferring benefit also shows physical formidability.

6. Experiment 5 – Conferring a benefit & size

We tested if the additional cue of physical size (the fact that the character giving the group a ball was bigger) was necessary for generating the prediction of continued obedience to a character who conferred benefit upon the group. Events were identical to those of Exp. 4 except that the character now had three balls (more resources to distribute) and was bigger (Fig. 1E and Movie-S8 to S10).

6.1. Methods

6.1.1. Participants

Participants were 16 full-term infants (8 female; $M_{Age} = 21$ mo, 8 d; age range = 19 mo, 10 d to 23 mo, 19 d) from Norwegian-speaking families. Another 10 infants were excluded because they looked the maximum time allowed in all the test pairs (2), because they were overly fussy or active (6) or distracted in test by an object item (1), or because of parental interference (1).

6.1.2. Design

Events were identical to those of Exp. 4 except that in the character-familiarization event the character had three colorful balls and was bigger, about twice the size of a protagonist (Fig. 1E and Movie-S8). The character was bigger also in the order-familiarization event (Movie-S9) and in the test events (see Movie-S10 and S11 for the obey and disobey events).

6.1.3. Procedure

The procedure was identical to that of Exp. 4. Interobserver agreement averaged 97 % per trial per infant.

6.2. Results

Again, we found positive Bayesian evidence that infants looked about equally at the disobey ($M=40.75$, $SD=12.16$) and obey ($M=41.43$, $SD=15.15$) events, $t(15) = 0.21$, $p = 0.834$, $d = 0.05$, $BF_{0+} = 3.32$ (Fig. 3).

6.3. Interim discussion

This result suggests that neither conferring benefit nor the formidability cue of physical size sufficed in generating the prediction of respect and continued obedience.

7. Experiment 6 – Receiving a tribute

We tested if receiving a tribute (the fact that the group voluntarily offered their ball to the character) sufficed in generating the prediction of continued obedience. Events were identical to those of Exp. 3 except that the character received a tribute from the group; one protagonist approached her, gave her the ball, and returned to its position (Fig. 1F and Movie-S12).

7.1. Methods

7.1.1. Participants

Participants were 16 full-term infants (8 female; $M_{Age} = 21$ mo, 2 d; age range = 19 mo, 4 d to 23 mo, 28 d) from Norwegian-speaking families. Another 4 infants were excluded because they were overly fussy or active.

7.1.2. Design

Events were identical to those of Exp. 4 except that in the character-familiarization event the character received a tribute from the group. To start, infants were presented with the initial sequence of the character-familiarization event used in Exp. 3 where the protagonists played with a ball and moved up-and-down once the character entered the scene. Next, one protagonist approached the character, gave her the ball, and returned to its position. The character left the scene with the ball and the protagonists formed a line (Fig. 1F and Movie-S12).

7.1.3. Procedure

The procedure was identical to that of Exp. 4 with the exception that all observations were recoded off-line. Interobserver agreement averaged 97 % per trial per infant.

7.2. Results

After the familiarization where a character received tribute, infants looked significantly longer at the obey ($M=42.99$, $SD=11.70$) than at the disobey ($M=36.98$, $SD=11.90$) event, $t(15) = 2.39$, $p = 0.031$, $d = 0.60$, $BF_{0-} = 4.33$ (Fig. 3).

7.3. Interim Discussion

This result suggests that infants expected the protagonists to disobey. We shall return on the explanation of this finding in the General Discussion.

8. Experiment 7 – Imposing a cost

We tested if imposing cost (the fact that the character forcefully took the ball from the group) sufficed in generating the expectation

of continued obedience. In the familiarization phase, infants saw the character approaching and taking the ball from one of the protagonists pushing it slightly (Fig. 1G and Movie-S13).

8.1. Methods

8.1.1. Participants

Participants were 16 full-term infants (6 female; $M_{Age} = 21$ mo, 13 d; age range = 19 mo, 2 d to 23 mo, 29 d) from Norwegian-speaking families. Another 4 infants were excluded because they were overly fussy or active (3) or distracted in test by food items (1).

8.1.2. Design

Events were identical to those of Exp. 6 except that in the character-familiarization event the character imposed a cost upon the protagonists. Instead of being approached by the group, she approached the protagonists, took the ball from one of them and pushed it slightly, returned to her position, and after a 3-s pause left the scene (Fig. 1G and Movie-S13).

8.1.3. Procedure

The procedure was identical to that of Exp. 4. Interobserver agreement averaged 97 % per trial per infant.

8.2. Results

We found positive Bayesian evidence that infants looked about equally at the disobey ($M=38.89$, $SD=10.26$) and obey ($M=40.43$, $SD=10.54$) events, $t(15) = 0.45$, $p = 0.658$, $d = 0.11$, $BF_{0+} = 5.29$ (Fig. 3).

8.3. Interim discussion

This result suggests that imposing cost does *not* generate the prediction of continued obedience in the just-linguistic infant mind.

9. Experiment 8 – hitting

We tested if violent aggression (the fact that the character hit the group with her stick) sufficed in generating the expectation of continued obedience. Infants were presented with a character-familiarization event where the character carried a stick, approached the protagonists and hit them on top of their head twice with the stick (see Fig. 1H, Movie-S14).

9.1. Methods

9.1.1. Participants

Participants were 16 full-term infants (7 female; $M_{Age} = 21$ mo, 2 d; age range = 19 mo, 11 d to 23 mo, 28 d) from Norwegian-speaking families. Another 8 infants were excluded because they looked the maximum time allowed in the test pairs (2), because they were overly fussy or active (3) or distracted in test by an object item (1), or because of bad internet connection (2).

9.1.2. Design

Events were identical to those of Exp. 7 except that in the character-familiarization event the character hit them (Fig. 1H). To start, the protagonists played ball like in Exp. 7. Next, the character entered the scene from the left and carried a stick; she then approached the protagonist who stood front left and hit it on top of the head twice with the stick. The protagonist winced at each blow and said, “Ouch!” (in Norwegian, “Au!”). After receiving two blows, the protagonist moved away from the character toward the house on the left of the scene, together with the protagonist who stood front right. The same sequence was repeated with the protagonist who stood at the back and had the ball; it also moved away, leaving the ball behind. The character returned to her initial position, together with the protagonists who then moved up-and-down before the character (Movie-S14).

9.1.3. Procedure

The procedure was identical to that of Exp. 4. Interobserver agreement averaged 97 % per trial per infant.

9.2. Results

Again, infants looked about equally at the disobey ($M=41.83$, $SD=8.15$) and obey ($M=43.94$, $SD=10.94$) events, $t(15) = 0.61$, $p = 0.554$, $d = 0.15$, $BF_{0+} = 5.77$ (Fig. 3).

9.3. Interim discussion

This result suggests that violent aggression does not generate the expectation of continued obedience.

10. General discussion

The power to direct the actions of others is a key aspect of leadership and the adaptive coordination benefits which it affords (Fiske, 1991; Glowacki & von Rueden, 2015; von Rueden & Van Vugt, 2015; Thomsen, 2020; Van Vugt & Ahuja, 2011). Here, we have demonstrated that just-linguistic infants precisely predict this directive aspect of leadership using a single, dedicated cue. Our findings yielded the following conclusions. First, three experiments (of which one replication and one control) demonstrate that the act of bowing alone suffices in generating the expectation of continued obedience that marks respect-based power and legitimate leadership. Second, at 21 months infants do *not* appear to use a number of other, theoretically plausible cues to generate such representations. In particular, neither conferring benefits nor imposing costs, receiving tribute, nor greater physical size generated the prediction of continued obedience to a character in her absence. Neither conferring benefits, imposing costs, nor receiving tribute thus appear to form part of the core understanding of respected leadership held by human infants, despite the central role given to these constructs in current theories and models of the evolution and the psychological foundations of leadership (Cheng et al., 2013; Durkee et al., 2020; Garfield et al., 2020; Hardy & Van Vugt, 2006; Henrich & Gil-White, 2001; Hooper et al., 2010; Van Vugt & von Rueden, 2020). By contrast, we have revealed that infants infer respect-based, directive power and predict continued obedience from the deference display of bowing/spontaneous prostration. That just-linguistic infants appear to focus exclusively on this overt, public deference display for predicting continued obedience is suggestive of the crucial role played by the directive, consensual social coordination ‘contract’ which it signifies – that is, a consensual, leader–follower relationship of respect (Fiske, 1992).

How can the development or presence of this early, relational ‘understanding’ be explained? Although bowing matches the respect and spontaneous deference displays observed across many human cultures and animal species, at their very young age and in the egalitarian Norwegian culture, the infants who participated in our research likely never saw anyone bow in their everyday life. This simple observation speaks against the hypothesis that the expectations we unveiled are outputs of gradual and extensive general learning occurring in the first year and a half of life, either through statistical learning mechanisms (Ruffman et al., 2012) or socialization processes with peers and/or adult figures (Killen & Smetana, 2015). Although social learning is undoubtedly central in acquiring culture-specific knowledge and norms, the present data are thus best, and most parsimoniously, explained by the presence in the human social mind of an abstract, early-developing core representation of respect-based, directive power and legitimate authority which takes being bowed before as dedicated cue (see Ting et al., 2020; Woo et al., 2022).

These findings show that physical dominance (forcefully taking the group’s ball) and violent aggression (hitting them with a stick) did *not* license the inference that the group would continue to obey the character issuing orders (thus replicating Margoni et al., 2018). Further, our findings (Exp. 4) show that conferring a benefit to a group does *not* license the inference that someone is a leader with directive power, not even when the character is also more physically formidable than the group members (Exp. 5). Finally, not only does receiving a tribute from a group not license the inference that the recipient is a leader, it actually generates the expectation that the group will *disobey* the recipient. Given the results of Experiments 4 and 5, this inference is unlikely driven by perceptions that the group consists of prestigious leaders because they conferred a benefit to the lone character, nor by perceptions that the group members need not obey the lone character because they are more formidable together. Instead, one might argue that infants could have interpreted the act of paying tribute to the leader as an exchange payment for a license to disobey its orders, thus possibly revealing an initial understanding of the logic of bribery (Reyes-Jaquez & Koenig, 2021, 2022).

The present demonstration that bowing is a sufficient cue for inferring respect-based leadership obviously does not imply that it is a necessary cue for doing so and that other relational information plays no role in predicting continued obedience in the infant mind. Note also that in many small-scale societies, generosity displays, if not properly accompanied by signs of humility, can signal *dominance* (fear-based power) rather than prestige (respect-based power) (Bliege Bird & Power, 2015; Lee, 1979; see also Boehm, 1999). Perhaps this observation may help explain why benefit conferment did not generate the expectation of continued obedience that marks respect-based power, despite the predictions from the competence-models of leadership (Cheng et al., 2013; Hardy & Van Vugt, 2006; Henrich & Gil-White, 2001; von Rueden et al., 2014). It is also possible that in the absence of overt deference cues, benefit conferment must occur in conjunction with overt acts of dominance and intimidation (such as forceful resource claiming and violent aggression) to result in predictions of continued obedience. Indeed, cues for dominance might also signal that the individual has at least the capacity to recruit relevant skills (e.g., fighting skills) to inflict costs on rival groups (Brand & Mesoudi, 2019). The ability to impose costs might be valued by the group, and so individuals who possess this ability may come to be respected as legitimate leaders, at least in situations such as war where this is important (Heck, 2023; Laustsen & Petersen, 2017; see also Hasty & Maner, 2023). However, because individuals might not be perceived as leaders if inferring costs is not followed by conferring benefits to the group, imposing a cost alone might not suffice in generating the representation of respected leadership. Future research should test if the *conjunction* of benefit conferment and cost infliction generates the expectation of continued obedience in the infant mind (or not). Alternatively, in principle it is also possible that benefit conferment did not generate the representation of leadership in the present study because the benefit was only temporarily loaned or perceived as too minimal and/or unrelated to critical, fundamental needs of the group members (e.g., providing water, food or shelter in a situation of extreme need). It is a wide-open empirical question whether these additional boundary conditions would be represented by infants and if they form necessary conditions for licensing inferences of leadership based on benefit conferral or, alternatively, whether only displays of deference license expectations of continued obedience in the infant mind.

The present demonstration that bowing alone suffices in generating the expectation that agents will continue to obey someone’s orders makes significant contributions to the emerging field of investigation of what are the core, early-developing expectations related to legitimate leadership (see Stavans & Baillargeon, 2019), raising many further questions regarding the core nature of leadership and providing the means for investigating them: Even though conferring benefits does not solicit expectations of continued obedience,

would infants expect a respected leader (that is, an agent for whom a group of subordinates bowed) to confer benefits to them? Would they expect a respected leader to refrain from imposing any cost on the group, or would they expect her to use force to get her way if others do not defer to her directions? If so, are respected leaders especially expected to enforce cooperation, and socio-moral and cultural norms, bearing the cost of punishing free-riders? And are subordinates expected to imitate (Bas et al., 2023), trust (e.g., Margoni et al., 2022) and learn from respected leaders, but not from individuals whose power stem from their formidability or the fear they instill (e.g., Fonn et al., 2022; Margoni et al., 2023)? Establishing the basic, universal and early-emerging representations of leader–follower and dominant-subordinate relations may shed light on how children use such a core ‘relational grammar’ (Thomsen & Carey, 2013) to rapidly develop culture-specific representations and attributions for navigating the hierarchical structures of their particular social world. Indeed, an emerging body of research with slightly older children is suggestive of both the universal and cultural specificity of the inferences and affordances of different forms of social rank. Whereas children differentiate between formidability-based dominance, competence-based prestige and subordination across cultures (e.g., Sequeira et al., 2024), studies have also revealed cultural differences in children’s expectations, leadership attributions, leadership identifications and learning preferences (Kajanus, 2023). For instance, early work found that preschoolers in Guatemala and France place selective epistemic trust in dominant over subordinate agents (Bernard et al., 2016; Castelain et al., 2016), but another study found the reverse effect in a relatively small Japanese sample (Charafeddine et al., 2019) and in egalitarian Norway, Fonn et al. (2022) found strong Bayesian evidence that preschoolers do not differentiate between dominant and subordinate characters in a word-learning paradigm across five experiments that also eliminated potential experimenter effects. Indeed, Sequeira et al. (2024) found no cross-cultural differences that Colombian, Finnish and US children prefer to learn from a prestigious rather than dominant character. Yet, school-age children in UK, but not in China, expect high-ranking individuals to prevail over subordinates in conflicts for resources (Kajanus et al., 2020); children in egalitarian Finland are more likely to attribute leadership to a prestigious agent than are children in Colombia or USA (Sequeira et al., 2024). Evidence of cross-cultural differences in how young children reason about power further highlights the crucial need of infant research to test evolutionary models of human social mind.

Last, in the present study where agents were ordered to go to bed, one might argue that infants could have interpreted the scene as exemplifying of a concrete parent–child relationship. Speaking against this possibility, however, is the fact that the only predictive cue for obedience used by the participating Norwegian infants was bowing, and no one bows before their parents in this cultural setting. Still, if the bowing-obedience expectations we have demonstrated here do indeed sprout from an abstract representation of leader–follower relations, then similar results should also be expected if the leader character issued an order about something else.

In conclusion, we have demonstrated that infants use the act of bowing as a dedicated, sufficient cue for respect-based power and consensual authority to give directions and issue orders that will be followed, even in one’s absence. This highlights the critical importance of leadership and the coordination benefits that it affords in human social life. Because it is unlikely that the participating Norwegian infants ever saw anyone bow in their everyday life, these results unveil a representation that forms part of a fundamental, dedicated and early-developing capacity for navigating leader–follower relations which was likely built by evolution.

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CRedit authorship contribution statement

Francesco Margoni: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lotte Thomsen:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Supplementary materials and raw data have been made publicly accessible at OSF and can be accessed at <https://osf.io/5mbpv/>.

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